AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM270
Artificial intelligence

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 15/Nov./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 .....is defined by how much memory is needed to perform the search.

A optimality
B Space complexity
C Time complexity
D Completeness

2 The moves that the chess piece is legally of making is called .....

A State space
B Symbol
C Process
D State

3 In expert systems, everything after the THEN in the IF-THEN rules is called .....

A premise
B inference
C consequent
D None of the above

4 ..... start with a possible solution and then play with it, checking if better solutions can be found by making small alterations at various points.

A Local search algorithm
B Exact algorithm
C Approximate algorithm
D Construction algorithm

5 _____ are both categorized as uninformed search algorithms.

A Iterative-deepening search and Beam search
B A* and Best-first search
C A* and Beam search
D Breadth-first search and Depth-first search

6 In .....a learning system is presented with the inputs and desired outputs of a program or procedure. The system has to derive the program that satisfies these constraints.

A inductive programming
B Supervised leaning
C Speed-up learning
D Unsupervised learning

7 In_____, there is a set of exemplars, each one accompanied by a teaching input that indicates the correct or the desired answer.

A Self-learning
B Supervised learning
C Automated learning
D Unsupervised learning

8 When we create a system that tries to mimic intelligence, it is called .. ..

A Simulation
B optimization
C emulation
D None of the above

9 In ..... We always start from hypothesis until reaching the goal

A Backward chaining
B Satisfaction problem
C Forward chaining
D None of the above

10 "The sky is clear" is an example of .....knowledge

A Procedural
B Declarative
C Base
D meta

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

a. Describe the α-β pruning minimax algorithm for evaluating nodes in a game tree. Explain the concept of alpha and beta cutoffs.[5 marks]

b. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	A	A	C	C	D	D	D	E	E	F	F	G	G	G	H	H	H	H	I	I	I	I	J	K	L	L	L	L
t	36	C	D	E	F	G	H	I	J	K	L	22	24	33	19	8	27	7	23	35	43	2	14	5	15	17	45	34	1	48

use minimax algorithm to evaluate each node in the tree.

Question 3: (15 marks)

a. Analyze the role of the agenda in STRIPS planning. How does the agenda manage the search process and ensure that progress is made towards achieving the goal conditions?

b. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	B	C	C	D	D	D	D	E	E	E	F	F	F
t	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P

Given the aforementioned tree, apply BFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is O. [10 marks]

Question 4: (15 marks)

a. Explain the rule used to calculate the evaporation of pheromones in the pheromone matrix. How does this rule affect the concentration of pheromones over time? [5 marks]

b. At a given point of time and given an ant, the ant found 5 possible routes at its journey. The pheromone amounts on each route are :

0.000000 1.000000 10.000000 10.000000 0.000000

If the length of each route which is inversely proportional to the route desirability is known to the ant and lengths are :

35.000000 60.000000 17.000000 58.000000 57.000000

Calculate the probability of selecting each route by that given ant

Assume Alpha=0.6 and beta= 0.0

End of questions

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